

Please attach this cover sheet to your completed homework

Homework 6 (Ch. 12.5 – 12.6)

ECE 311 – Hang Gao

Name: Gianni Avilan-Losee

Problem	Course outcome	Grade
12.17	1.c.	/20
12.18 (a)	2.c.	/10
12.18 (b)	1.c.	/10
12.18 (c)	1.c.	/10
12.18 (d)	1.c.	/10
12.19 (a)	2.c.	/10
12.19 (b)	1.c.	/10
12.22	1.c.	/20
TOTAL:		/100

ABET Assessed Outcomes

1.c. Use appropriate power system models to formulate solutions.

Please attach this cover sheet to your completed homework

2.c. Analyze power systems with safety, social, environmental, and economic factors and constraints.

Homework #6

Due date: Sunday, March 23rd by 11:59 PM (Electronically – Canvas)

12.17 A synchronous generator connected directly to an infinite bus is generating at its maximum real power. The line-to-line voltage of the infinite bus is 480 V and the synchronous reactance of the generator is $5\ \Omega$. Compute the reactive power of the generator.

12.18 A 1 GW synchronous generator is connected to an infinite bus through a transmission line. The synchronous reactance of the generator is $9\ \Omega$ and the inductive reactance of the transmission line is $3\ \Omega$. The infinite bus voltage is 110 kV. If the generator delivers its rated power at unity power factor to the infinite bus, compute the following:

- Terminal voltage of the generator
- Equivalent field voltage
- Real power output at the generator terminals
- Reactive power output at the generator terminals

12.19 A synchronous generator is connected directly to an infinite bus. The voltage of the infinite bus is 15 kV. The excitation of the generator is adjusted until the equivalent field voltage E_f is 14 kV. The synchronous reactance of the machine is $5\ \Omega$. Compute the following:

- Pullover power (maximum power)
- Equivalent excitation voltage that increases the pullover power by 20%

12.22 A synchronous motor excited by a 5 kV source has a synchronous reactance of $5\ \Omega$. The motor is running without any mechanical load (i.e., the output real power is zero). Ignore all losses and calculate the equivalent field voltage E_f that delivers 3 MVar to the source. Also, draw the phasor diagram.

12.17 Using the power angle equation for reactive power

$$Q = 3 \frac{V_o}{X_s} (E_f \cos \delta - V_o),$$

and assuming that $\delta = 90^\circ$, since the generator is operating at its maximum power, solve for the reactive power Q as

$$Q = 3 \frac{277.13}{5} (-277.13) = -46.081 \text{ kVAR}.$$

12.18 a) Terminal voltage may be found as a function of the transmission line current, which itself is a function of the power angle δ . Finding power angle δ and effective field voltage E_f as a function of P and Q ,

$$P = 1 \times 10^9 = \frac{3V_o E_f}{X} \sin \delta \rightarrow E_f \sin(\delta) = \frac{PX}{3V_o} = 62.984 \text{ kV}$$

$$Q = 0 = \frac{3V_o}{X_s} (E_f \cos(\delta) - V_o) \rightarrow E_f \cos(\delta) = V_o = 63.51 \text{ kV}.$$

Therefore,

$$\overline{E}_f = E_f \cos(\delta) + jE_f \sin(\delta) = 63.51 + j62.98 \text{ kV},$$

and

$$\delta = 44.76^\circ.$$

Then, we may find current as

$$\overline{I}_a = \frac{\overline{E}_f - \overline{V}_o}{\overline{X}} = \frac{89.44 \angle 44.76^\circ - 63.51 \angle 0^\circ}{12 \angle 90^\circ} = 5.25 \angle 0^\circ \text{ kA}.$$

Then, finding line voltage,

$$\overline{V}_l = \overline{V}_o + \overline{I}_a \overline{X}_l = \frac{110}{\sqrt{3}} \angle 0^\circ + ([5.25 \angle 0^\circ] \times [3 \angle 90^\circ]) = 65.43 \angle 13.93^\circ \text{ kV}.$$

Finally, line-to-line terminal voltage may be found as

$$65.43 \times \sqrt{3} = 113.33 \text{ kV}.$$

b) Equivalent field voltage, as shown in the previous step, is

$$E_f = |\overline{E}_f| = 89.44 \text{ kV}.$$

c) Real power output at the generator terminals is P , that is,

$$P_t = P = 1 \text{ GW}.$$

d) Reactive power output at the terminals may be found using α , the angle between terminal voltage and $\overline{E}_f$, as

$$Q_t = \frac{3V_t}{X_s} (E_f \cos(\alpha) - V_t) = \frac{3(65.43)}{9} (89.44 \cos(30.83^\circ) - 65.43) = 248 \text{ MVAR}.$$

12.19 a) Pullover power occurs when $\delta = 90^\circ$, and so

$$P_{\max} = \frac{3V_o E_f}{X_s} = \frac{3 \times \frac{15}{\sqrt{3}} \times 14}{5} = 72.75 \text{ MW}.$$

b) To increase $P_{\max}$ to 87.3^{MW} using a higher excitation voltage, solving for E_f ,

$$E_f = \frac{P_{\max} X_s}{3V_o} = \frac{87.3 \times 5}{3 \times \frac{15}{\sqrt{3}}} = 16.8^{\text{kV}}.$$

12.22 Reactive power consumed by a motor is given by

$$Q = \frac{V_t}{X_s} (V_t - E_f \cos(\delta)),$$

and so, using $Q = -3^{\text{MVAr}}$ and $\delta = 0^\circ$,

$$E_f = -\frac{Q X_s}{V_t} + V_t = \frac{3 \times 5}{5} + 5 = 8^{\text{kV}}.$$